

AI2: Intro to Uncertainty Quantification

Ata Kabán

School of Computer Science
University of Birmingham



UNIVERSITY OF
BIRMINGHAM

Overview

In this lecture are interested in the following questions:

- How reliable is my estimator?
- How to estimate the reducible component of uncertainty (in any model)?

We learn:

- Bias-variance analysis of estimators
- An empirical method to quantify *epistemic uncertainty* from the data

Part 1:

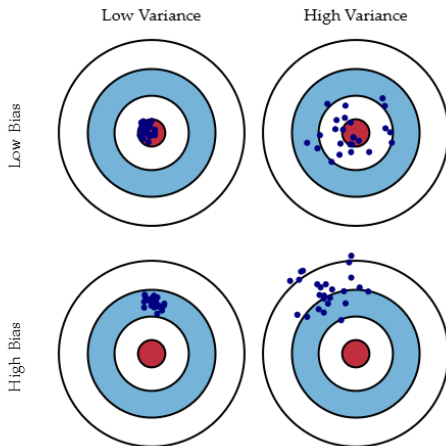
Bias-variance analysis of estimators

- 'Estimation' and 'inference' are often used interchangeably.
- Goal: Construct an estimator $\hat{\theta}_n$ for the unknown parameter θ^* given observed data S . The index n is to indicate that we obtained this estimate from n examples.
- We briefly talked about bias and variance of the expected square error of MLE last week; we return to these concepts for better understanding and wider applicability.
- Example: Estimate the parameter of a Bernoulli distribution from coin flips.
- Note: $\hat{\theta}_n$ is a random variable due to its dependence on the finite and possibly noisy dataset S .

Intuitions

Imagine many training sets of size n generated from the some unknown distribution. From each we estimate a separate $\hat{\theta}_n$.

Question: How does the error change for different training sets?



The center of the red circle depicts the true parameter θ^* , the blue dots depict our estimates $\hat{\theta}_n$ obtained from each of the many training sets.

<https://scott.fortmann-roe.com/docs/BiasVariance.html>

Definitions

Suppose we have many training sets of size n generated from the some unknown distribution. From each we estimate a separate $\hat{\theta}_n$.

Question: How does the error change for different training sets?

Bias of an Estimator:

- Definition: Bias of $\hat{\theta}_n$ is $\text{Bias}(\hat{\theta}_n) = \mathbb{E}[\hat{\theta}_n] - \theta^*$.
- An estimator is **unbiased** if $\mathbb{E}[\hat{\theta}_n] = \theta^*$ for all θ^* .

Variance of an Estimator:

- Definition: Variance of $\hat{\theta}_n$ is $\text{Var}(\hat{\theta}_n) = \mathbb{E}[(\hat{\theta} - \theta^*)^2]$ (if $\hat{\theta}$ is a scalar variable) or $\text{Cov}[\hat{\theta}_n]$ (if $\hat{\theta}$ is a vector variable).
- Unlike bias, the variance does not directly depend on the true parameter θ^* .

Bias-Variance decomposition of the Mean Squared Error (this slide is from last week's lecture)

We look at the expected square error (expectation over the distribution that generated the training sets) from the true parameter value θ^* :

$$\mathbb{E}[(\hat{\theta} - \theta^*)^2] = \mathbb{E}[\hat{\theta}^2] - 2\mathbb{E}[\hat{\theta}]\theta^* + (\theta^*)^2$$

Add & subtract $\mathbb{E}[\hat{\theta}]^2$ to complete the square:

$$= \mathbb{E}[\hat{\theta}^2] + (\theta^* - \mathbb{E}[\hat{\theta}])^2 - \mathbb{E}[\hat{\theta}]^2$$

Rearrange to conclude:

$$\mathbb{E}[(\hat{\theta} - \theta^*)^2] = \underbrace{(\theta^* - \mathbb{E}[\hat{\theta}])^2}_{\text{Bias}[\hat{\theta}]^2} + \underbrace{\mathbb{E}[(\hat{\theta} - \mathbb{E}[\hat{\theta}])^2]}_{\text{Var}[\hat{\theta}]}$$

Worked Example: *Analysis of the ML and MAP estimators of the Bernoulli parameter*

- Recall the Bernoulli distribution is given by:

$$P(X = x|\theta) = \theta^x(1 - \theta)^{1-x}$$

where:

- X is a Bernoulli trial outcome ($X = 1$ for success, $X = 0$ for failure),
 - $\theta = \theta^* = P(X = 1|\theta)$ is the unknown true probability of success.
- We know 3 different estimators of θ from sequence of n independent trials (e.g. coin flips), $X_1, X_2, \dots, X_n$ drawn i.i.d. from $\text{Bernoulli}(\theta)$.

MLE and MAP estimators (derived last week)

The Maximum Likelihood Estimator for θ is:

$$\hat{\theta}_{MLE} = \frac{\sum_{i=1}^n X_i}{n}$$

The MLE is the sample mean of Bernoulli trials.

The Maximum a Posteriori (MAP) Estimator for θ is:

$$\hat{\theta}_{MAP} = \frac{\sum_{i=1}^n X_i + \alpha - 1}{n + \alpha + \beta - 2}$$

The MAP is the maximiser of the posterior distribution of θ when the prior distribution is $p(\theta) = \text{Beta}(\alpha, \beta)$.

Note: Bias-variance is a frequentist concept. While MAP (and Bayesian mean) derive from Bayesian framework, all estimators can be analysed in the frequentist framework.

Bias of the Estimators

Bias is defined as:

$$\text{Bias}(\hat{\theta}) = E[\hat{\theta}] - \theta$$

Bias of MLE:

$$\text{Bias}(\hat{\theta}_{MLE}) = E[\hat{\theta}_{MLE}] - \theta = E\left[\frac{\sum_{i=1}^n X_i}{n}\right] - \theta \quad (1)$$

$$= \frac{E[\sum_{i=1}^n X_i]}{n} - \theta = \frac{n\theta}{n} - \theta = \theta - \theta = 0 \quad (2)$$

Thus, the MLE is unbiased.

Bias of MAP:

$$\text{Bias}(\hat{\theta}_{MAP}) = E[\hat{\theta}_{MAP}] - \theta = \frac{n\theta + \alpha - 1}{n + \alpha + \beta - 2} - \theta$$

Thus, the MAP estimator is biased towards the prior mean, especially for small n , but becomes unbiased as n grows large.

Variance of the Estimators

Variance is given by:

$$\text{Var}(\hat{\theta}) = E[\hat{\theta}^2] - (E[\hat{\theta}])^2$$

Variance of MLE:

$$\text{Var}(\hat{\theta}_{MLE}) = \text{Var}\left(\frac{\sum_{i=1}^n X_i}{n}\right) = \frac{\text{Var}(\sum_{i=1}^n X_i)}{n^2} = \frac{\theta(1-\theta)}{n}$$

Variance of MAP Estimator:

$$\text{Var}(\hat{\theta}_{MAP}) = \frac{1}{(n + \alpha + \beta - 2)^2} \cdot \text{Var}\left(\sum_{i=1}^n X_i\right) = \frac{n\theta(1-\theta)}{(n + \alpha + \beta - 2)^2}$$

This can be made smaller than and $\text{Var}(\hat{\theta}_{MLE})$ by an informative prior (α, β away from 1). **Notice the tradeoff with $\text{Bias}(\hat{\theta}_{MAP})$.**

For large n , this becomes approx. $\text{Var}(\hat{\theta}_{MAP}) \approx \frac{\theta(1-\theta)}{n+\alpha+\beta-2}$. Similar to the MLE, but with a denominator that includes prior information (α and β).

Implications of Bias-Variance Analysis

- We want estimators that have low bias and low variance – but both are not achievable simultaneously with a finite sample, and there are tradeoffs.
- Bias-variance properties of estimators can guide the choice of estimator to use. MLE has low bias if n is sufficiently large, but it has high variance.
- High bias, low variance estimators (e.g., regularization methods – also interpretable as MAP estimators) improve stability and generalization.
- Bayesian estimators incorporate prior information, introducing bias but reducing variance. The Bayesian posterior mean is often biased but achieves lower MSE than frequentist estimators.
- The best choice depends on sample size, prior knowledge, and application needs.

Bias-variance analysis in more general

Bias-variance analysis is well suited whenever the square-error is used – e.g. in regression. We don't do this here, you will learn about it in the Machine Learning module in Year3. Here we just give the intuition very briefly

- Let S be a training dataset sampled from some distribution.
- Let $x \in \mathcal{X}$ denote an input and $y \in \mathbb{R}$ the corresponding output.
- Assume the true relationship is given by

$$y = f(x) + \epsilon,$$

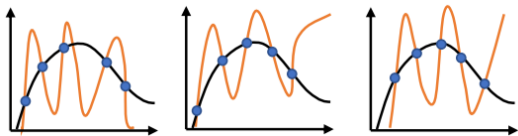
where ϵ is random noise with $\mathbb{E}[\epsilon] = 0$ and variance $\text{Var}(\epsilon) = \sigma^2$.

- A learning algorithm produces an estimator $\hat{f}(x; S)$.

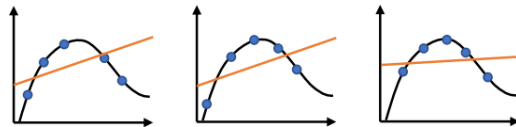
Bias-variance analysis in more general

Imagine many training sets of size n generated from the some unknown distribution. From each we estimate a separate predictor $\hat{f}_n$.

Question: Fix test point x . How does the error of $\hat{f}_n$'s from different training sets vary?



The learned $\hat{f}_n$'s look different each time! The training data is fitted well, but the true function is fitted poorly.



The learned function looks similar; Both the training data and the true function are fitted poorly.

Hint: We can be in one of 4 cases just like before. And it depends on our model choice! (in addition to the estimator used to train the parameters!)

Upper row: High variance, low bias. **Lower row:** Low variance, high bias.

Part 2:

Bootstrap for Quantifying Epistemic Uncertainty

- Bootstrapping ensembles repeatedly resample the training set, train a separate model on each sample, and average their predictions. This can reduce variance.
- In fact, bootstrapping ensembles can also be used as a general method for quantifying the epistemic uncertainty! - Let's have a look how this works.
 - ▶ Recall the Aleatoric and Epistemic Uncertainty decomposition.
 - ▶ We were able to decouple them analytically in linear regression, but in more complex models analytic methods become infeasible.
 - ▶ If you have an unbiased model, you can use bootstrap to estimate the epistemic uncertainty in practice.

Bootstrap for Estimating Epistemic Uncertainty

- Recall the predictive uncertainty can be written as a sum of aleatoric and epistemic uncertainty

$$y - \hat{f}(x; S) = y - f_{\hat{\theta}}(x) = \underbrace{y - f_{\theta^*}(x)}_{\text{Aleatoric uncertainty}} + \underbrace{f_{\theta^*}(x) - f_{\hat{\theta}}(x)}_{\text{Epistemic uncertainty}}$$

where θ^* are the true parameters of the function we try to learn.

- Hard to disentangle the two uncertainties of different nature:
 - ▶ We directly observe the predictive uncertainty $y - \hat{f}(x; S)$
 - ▶ But θ^* is unknown (even if we assume we know the general form of f)

Note. These notations interchangeably denote a trained model: $\hat{f}(x; S) = f_{\hat{\theta}}(x) = \hat{f}(x)$.

Epistemic Uncertainty

- The epistemic uncertainty is:

$$\text{Epistemic}(x) = f_{\theta^*}(x) - \hat{f}(x)$$

- ▶ Note, $\text{Epistemic}(x)$ is a random function of the random sample S .
 - ▶ Thus, $\text{Epistemic}(x)$ is itself a random variable.
- Goal: Estimate the distribution of $\text{Epistemic}(x)$
- **Assumption: Our model is unbiased** (i.e. we need to assume its Bias is zero):

$$\mathbb{E}_S[f_{\hat{\theta}}(x)] = f_{\theta^*}(x).$$

- But $S \sim_{i.i.d} \mathcal{D}$ but $\mathcal{D}$ is an unknown distribution :- (

What if we know $\mathcal{D}$?

- We could take k (a very large number of) samples of size n , each i.i.d. from $\mathcal{D}$, train the model on each
- Then $\{f_{\hat{\theta}}^{(i)}(x) - f_{\theta^*}(x)\}_{i=1}^k$ are i.i.d. samples from $\text{Epistemic}(x)$.
- By our unbiasedness assumption:

$$f_{\theta^*}(x) = \mathbb{E}_S[f_{\hat{\theta}}(x)] \approx \frac{1}{k} \sum_{i=1}^k f_{\hat{\theta}}^{(i)}(x) = \hat{\mu}(x)$$

- $\{f_{\hat{\theta}}^{(i)}(x) - \hat{\mu}(x)\}_{i=1}^k$ are ...approximately... i.i.d. samples from $\text{Epistemic}(x)$.
- *Problem: We don't have access to unlimited number samples from $\mathcal{D}$, but only have a single sample, S .*

Bootstrap

- Idea: Given samples $x_1, \dots, x_n \sim \mathcal{D}$, we can approximate the unknown probability distribution $\mathcal{D}$ by (here, $\delta(x - x_i)$ denotes a Dirac delta centered on x_i):

$$P(x) = \Pr(X = x) \approx \frac{1}{n} \sum_{i=1}^n \delta(x - x_i) = \hat{P}(x)$$

- This ...can be made to work... for continuous distributions using the probability density function:

$$p(x) \approx \hat{p}(x) = \frac{1}{n} \sum_{i=1}^n \delta(x - x_i) = \hat{p}(x)$$

- Now use this in place of the unknown $\mathcal{D}$.

Practical Algorithm for Estimating Epistemic Uncertainty

Algorithm **Bootstrap**(sample $S = \{(x_i, y_i)\}_{i=1}^n$)

- Define empirical distribution:

$$\hat{p}(x) = \frac{1}{n} \sum_{i=1}^n \delta(x - x_i)$$

- Generate k bootstrap datasets $S_1, \dots, S_k \sim \hat{P}$
- Train models on each dataset: $\hat{f}^{(i)} = \text{train}(S_i)$
- **Return** epistemic uncertainty estimate:

$$U(x) = \{\hat{f}^{(i)}(x) - \hat{\mu}(x)\}_{i=1}^k,$$

where $\hat{\mu}(x) = \frac{1}{k} \sum_{i=1}^k \hat{f}^{(i)}(x)$

Example Application to Active Learning

- Train bootstrapped ensemble of models $\hat{f}^{(i)}(x)$
- Query the label for the example where the ensemble has the highest disagreement:

$$x^* = \arg \max_x \frac{1}{k^2} \sum_{i,j=1}^k I[\hat{f}^{(i)}(x) \neq \hat{f}^{(j)}(x)]$$

- Other metrics based on epistemic uncertainty can also be used
- Also commonly used for guiding exploration in reinforcement learning
- More generally, decision-making with opportunity to gather information

Conclusions

We learned about

- bias-variance decomposition as a method of analysis
 - ▶ in estimation problems
 - ▶ in regression problems
- the relationship between bias-variance and underfitting-overfitting in regression
- methods used in practice to balance bias-variance or underfitting-overfitting: regularisation, and bootstrap (brief intro)
- a bootstrap algorithm to estimate epistemic uncertainty from the data